

MY SQL

-- 1. Create Table

```
mysql> CREATE TABLE student_marks1 (  
->     roll_no INT PRIMARY KEY,  
->     name VARCHAR(50),  
->     subject VARCHAR(50),  
->     marks DECIMAL(5,2)  
-> );  
Query OK, 0 rows affected (0.08 sec)
```

-- 2. Insert Data

```
mysql> INSERT INTO student_marks1 (roll_no, name, subject, marks) VALUES  
-> (1, 'Ravi', 'Math', 85.50),  
-> (2, 'Sita', 'Math', 92.75),  
-> (3, 'Anil', 'Math', 78.40),  
-> (4, 'Priya', 'Math', 88.90),  
-> (5, 'Vijay', 'Math', 80.25),  
-> (6, 'subbusir', 'aws', 98.50),  
-> (7, 'dwaraka', 'dbms', 95.75),  
-> (8, 'ranjani', 'english', 97.40),  
-> (9, 'kavithamam', 'aws1', 99.90),  
-> (10, 'seetha', 'azure', 82.25);  
Query OK, 10 rows affected (0.02 sec)  
Records: 10  Duplicates: 0  Warnings: 0
```

-- 3. Simple Select

```
mysql> SELECT * FROM student_marks1;  
+-----+-----+-----+-----+  
| roll_no | name      | subject | marks |  
+-----+-----+-----+-----+  
| 1       | Ravi      | Math    | 85.50 |  
| 2       | Sita      | Math    | 92.75 |  
| 3       | Anil      | Math    | 78.40 |  
| 4       | Priya     | Math    | 88.90 |  
| 5       | Vijay     | Math    | 80.25 |  
| 6       | subbusir  | aws     | 98.50 |  
| 7       | dwaraka   | dbms    | 95.75 |  
| 8       | ranjani   | english | 97.40 |  
| 9       | kavithamam | aws1    | 99.90 |  
| 10      | seetha    | azure   | 82.25 |  
+-----+-----+-----+-----+  
10 rows in set (0.01 sec)
```

-- 4. Aggregate Functions

```
mysql> SELECT COUNT(*) AS total_students FROM student_marks1;
+-----+
| total_students |
+-----+
|          10    |
+-----+
1 row in set (0.02 sec)
```

```
mysql> SELECT SUM(marks) FROM student_marks1;
+-----+
| SUM(marks) |
+-----+
|    899.60   |
+-----+
1 row in set (0.01 sec)
```

```
mysql> SELECT AVG(marks) FROM student_marks1;
+-----+
| AVG(marks) |
+-----+
|  89.960000 |
+-----+
1 row in set (0.00 sec)
```

```
mysql> SELECT MAX(marks) FROM student_marks1;
+-----+
| MAX(marks) |
+-----+
|    99.90   |
+-----+
1 row in set (0.00 sec)
```

```
mysql> SELECT MIN(marks) FROM student_marks1;
+-----+
| MIN(marks) |
+-----+
|    78.40   |
+-----+
1 row in set (0.00 sec)
```

-- 5. Filtering with WHERE

```
mysql> SELECT * FROM student_marks1 WHERE marks > 85;
```

roll_no	name	subject	marks
1	Ravi	Math	85.50
2	Sita	Math	92.75
4	Priya	Math	88.90
6	subbusir	aws	98.50
7	dwaraka	dbms	95.75
8	ranjani	english	97.40
9	kavithamam	aws1	99.90

7 rows in set (0.00 sec)

```
mysql> SELECT * FROM student_marks1 WHERE marks >= 90;
```

roll_no	name	subject	marks
2	Sita	Math	92.75
6	subbusir	aws	98.50
7	dwaraka	dbms	95.75
8	ranjani	english	97.40
9	kavithamam	aws1	99.90

5 rows in set (0.00 sec)

```
mysql> SELECT * FROM student_marks1 WHERE marks < 80;
```

roll_no	name	subject	marks
3	Anil	Math	78.40

1 row in set (0.00 sec)

```
mysql> SELECT * FROM student_marks1 WHERE name LIKE 'P%';
```

roll_no	name	subject	marks
4	Priya	Math	88.90

1 row in set (0.00 sec)

```
mysql> SELECT * FROM student_marks1 WHERE name IN ('Ravi', 'Sita', 'Vijay');
```

roll_no	name	subject	marks
1	Ravi	Math	85.50
2	Sita	Math	92.75
5	Vijay	Math	80.25

3 rows in set (0.01 sec)

```
mysql> SELECT * FROM student_marks1 WHERE marks > 85 AND (subject = 'Math' OR name LIKE 'P%');
```

roll_no	name	subject	marks
1	Ravi	Math	85.50
2	Sita	Math	92.75
4	Priya	Math	88.90

3 rows in set (0.00 sec)

-- 6. Update Queries

```
mysql> UPDATE student_marks1 SET marks = 90.00 WHERE roll_no = 3;
Query OK, 1 row affected (0.02 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

```
mysql> UPDATE student_marks1 SET subject = 'Remedial Math' WHERE marks < 80;
Query OK, 0 rows affected (0.00 sec)
Rows matched: 0  Changed: 0  Warnings: 0
```

-- 7. Delete Queries

```
mysql> DELETE FROM student_marks1 WHERE roll_no = 5;
Query OK, 1 row affected (0.02 sec)
```

```
mysql> DELETE FROM student_marks1 WHERE marks < 75;
Query OK, 0 rows affected (0.00 sec)
```

-- 8. Sorting

```
mysql> SELECT * FROM student_marks1 ORDER BY marks ASC;
+-----+-----+-----+-----+
| roll_no | name   | subject | marks |
+-----+-----+-----+-----+
|      10 | seetha | azure   | 82.25 |
|       1 | Ravi   | Math    | 85.50 |
|       4 | Priya  | Math    | 88.90 |
|       3 | Anil   | Math    | 90.00 |
|       2 | Sita   | Math    | 92.75 |
|       7 | dwaraka | dbms    | 95.75 |
|       8 | ranjani | english | 97.40 |
|       6 | subbusir | aws     | 98.50 |
|       9 | kavithamam | aws1    | 99.90 |
+-----+-----+-----+-----+
9 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM student_marks1 ORDER BY marks DESC;
+-----+-----+-----+-----+
| roll_no | name   | subject | marks |
+-----+-----+-----+-----+
|       9 | kavithamam | aws1    | 99.90 |
|       6 | subbusir   | aws     | 98.50 |
|       8 | ranjani    | english | 97.40 |
|       7 | dwaraka    | dbms    | 95.75 |
|       2 | Sita       | Math    | 92.75 |
|       3 | Anil       | Math    | 90.00 |
|       4 | Priya      | Math    | 88.90 |
|       1 | Ravi       | Math    | 85.50 |
|      10 | seetha     | azure   | 82.25 |
+-----+-----+-----+-----+
9 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM student_marks1 ORDER BY name ASC;
```

roll_no	name	subject	marks
3	Anil	Math	90.00
7	dwaraka	dbms	95.75
9	kavithamam	aws1	99.90
4	Priya	Math	88.90
8	ranjani	english	97.40
1	Ravi	Math	85.50
10	seetha	azure	82.25
2	Sita	Math	92.75
6	subbusir	aws	98.50

```
9 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM student_marks1 ORDER BY name DESC;
```

roll_no	name	subject	marks
6	subbusir	aws	98.50
2	Sita	Math	92.75
10	seetha	azure	82.25
1	Ravi	Math	85.50
8	ranjani	english	97.40
4	Priya	Math	88.90
9	kavithamam	aws1	99.90
7	dwaraka	dbms	95.75
3	Anil	Math	90.00

```
9 rows in set (0.00 sec)
```

-- 9. Group By

```
mysql> SELECT subject, SUM(marks) AS total_marks FROM student_marks1 GROUP BY subject;
```

subject	total_marks
Math	357.15
aws	98.50
dbms	95.75
english	97.40
aws1	99.90
azure	82.25

```
6 rows in set (0.01 sec)
```

```
mysql> SELECT subject, AVG(marks) AS avg_marks FROM student_marks1 GROUP BY subject;
```

subject	avg_marks
Math	89.287500
aws	98.500000
dbms	95.750000
english	97.400000
aws1	99.900000
azure	82.250000

```
6 rows in set (0.00 sec)
```

```
mysql> SELECT subject, COUNT(*) AS num_students FROM student_marks1 GROUP BY
subject;
+-----+-----+
| subject | num_students |
+-----+-----+
| Math    | 4            |
| aws     | 1            |
| dbms    | 1            |
| english | 1            |
| aws1    | 1            |
| azure   | 1            |
+-----+-----+
6 rows in set (0.00 sec)
```

-- 10. Having

```
mysql> SELECT subject, AVG(marks) AS avg_marks
-> FROM student_marks1
-> GROUP BY subject
-> HAVING AVG(marks) > 90;
+-----+-----+
| subject | avg_marks |
+-----+-----+
| aws     | 98.500000 |
| dbms    | 95.750000 |
| english | 97.400000 |
| aws1    | 99.900000 |
+-----+-----+
4 rows in set (0.00 sec)
```

```
mysql> SELECT subject, COUNT(*) AS num_students
-> FROM student_marks1
-> GROUP BY subject
-> HAVING COUNT(*) > 1;
+-----+-----+
| subject | num_students |
+-----+-----+
| Math    | 4            |
+-----+-----+
1 row in set (0.00 sec)
```

```
mysql> ALTER TABLE bank_transactions
-> ADD account_no VARCHAR(20);
Query OK, 0 rows affected (0.14 sec)
Records: 0 Duplicates: 0 Warnings: 0
```